ML-DSA.KeyGen()

Generates a public-private key pair.

Output: Public key $pk \in \mathbb{B}^{32+32k(\mathsf{bitlen}\ (q-1)-d)}$
and private key $sk \in \mathbb{B}^{32+32+64+32\cdot((\ell+k)\cdot\mathsf{bitlen}\ (2\eta)+dk)}$.

1: $\xi \leftarrow \mathbb{B}^{32}$ ▷ choose random seed
2: **if** $\xi =$ NULL **then**
3: **return** $\perp$ ▷ return an error indication if random bit generation failed
4: **end if**
5: **return** ML-DSA.KeyGen_internal (ξ)